

# Sushil Duseja

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## Summary

Senior technology and governance leader with 19 years spanning tech and financial services, building compliance infrastructure, control frameworks, and operating models in highly regulated, high-volume environments. Brings a strong foundation for **AI safety**, oversight, and responsible enterprise adoption.

Specialized in legacy-to-cloud migrations and building secure, scalable ecosystems that fuel business growth and **AI-driven innovation** for leading global institutions.

Experienced in shaping architecture from complex business needs, with working knowledge of **AI-led design** and the supporting stack, helping organizations open new revenue paths, maintain stability in changing markets, and drive digital execution.

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## Technical Expertise

- **Java/Spring Boot, Kafka, Solace:** built Commerzbank's risk and position-keeping service as the single source of truth for \$500M+ daily volume across MiFID II and Dodd-Frank reporting workflows.
  - **FIX Protocol, multicast, KDB+, MongoDB:** enhanced Commerzbank's market data layer; execution latency dropped 15% and throughput improved 10% for high-frequency algorithmic strategies.
  - **AWS, Docker, Kubernetes, GitOps:** led Barclays' £6M cloud-native migration for FX systems handling \$1B+ daily volume; release cycles shortened 80% and uptime held at 99.97%.
  - **Next.js, GraphQL, microservices:** designed the API and frontend layer for Barclays' modernized FX platform; trading and compliance surfaces consumed real-time data through a unified GraphQL contract.
  - **OWASP, DevSecOps, CI/CD automation, ISO/IEC 27001:** embedded security controls in Barclays' delivery pipeline; vulnerability count dropped 40% with zero findings across regulatory examinations.
  - **PostgreSQL, Oracle, MongoDB, KDB+:** delivered the data foundation at Barclays and Commerzbank; lineage and validation improvements at Commerzbank lifted P&L reporting accuracy by 25%.
  - **Python, TypeScript, Streamlit, Transformers, fairness metrics libraries:** built AI safety and governance prototypes covering prompt injection defense, fairness auditing, and multi-agent trade surveillance.
  - **NIST CSF:** Applied NIST Cybersecurity Framework as the dual governance structure for platform controls, mapping technical implementation to MiFID II regulatory requirements and cybersecurity GRC obligations in parallel.
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## Research Outputs / Preprints

Duseja, S. (2026). Beyond Remediation: An Irreversibility Threshold for Preventive AI Governance. SSRN (DOI:). <https://dx.doi.org/10.2139/ssrn.6384060>. Also available on Zenodo. [10.5281/zenodo.18908588](https://zenodo.org/record/105281/10.5281/zenodo.18908588).

Routes AI harms along four tracks (standard remediation, strict liability, precautionary technical architecture, macroeconomic transition policy) based on restorability within governance-relevant time horizons. Grounded in comparative analysis of the EU AI Act, NIST AI RMF, and UK White Paper.

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## Professional Experience

### Barclays | Vice President, Platform Modernization | Singapore | March 2023 – March 2025

Led £6M modernization of FX trading systems handling \$1B+ daily volume across 100+ institutional clients.

- Built a real-time data pipeline processing 10M+ market events daily; the pipeline powered an RFQ pricing feature that surfaced intelligent pricing signals to traders at point of decision, the first ML-adjacent capability on the platform.
- Designed behavior logging and data lineage tracing across microservices, producing a continuous audit trail of system state changes used for both regulatory examination and anomaly investigation.
- Implemented circuit breakers and threshold-based escalation controls on automated trading workflows, routing edge-case decisions to human reviewers rather than defaulting to system-side resolution.
- Applied NIST frameworks as the governance structure for platform controls, mapping technical implementation to regulatory reporting requirements under MiFID II.
- Embedded DevSecOps practices across the CI/CD pipeline; vulnerability count dropped 40% with zero findings across regulatory examinations.
- Automated control evidence collection so compliance staff reviewed exceptions rather than assembled artifacts, cutting audit preparation overhead significantly.

### Commerzbank | Vice President, Delivery Lead – Algo Trading Solutions | Singapore September 2017 – July 2022

Led 13-engineer team building algorithmic trading systems processing \$500M+ daily volume under MiFID II and Dodd-Frank.

- Engineered real-time position limit enforcement as a hard programmatic constraint on the automated trading system; breaches triggered immediate escalation to the risk function rather than system-side override.
- Built anomaly thresholds and kill-switch controls into the algo execution layer; out-of-bound signals halted automated decisions and queued them for human review before re-execution.
- Built the risk and position-keeping service in Java/Spring Boot as the single source of truth for risk analytics; P&L reporting accuracy improved 25% and the service fed \$1B+ risk exposure monitoring across all algorithmic strategies.
- Improved the market data layer using FIX Protocol, KDB+, and MongoDB; throughput improved 10%, execution latency dropped 15%, processing 100K+ orders daily at low millisecond speeds.
- Led TFX T-Clear integration enabling Japan market entry, capturing \$10M+ annual revenue; defined system interfaces across Java backends and high-throughput messaging queues.
- Delivered multiple platform initiatives directly attributed to 15% revenue growth across the algo trading book.

### IRIS Software Inc. | Associate Project Manager | India | 2017 – 2017

- **Client:** Bank of Montreal - DTCC trade confirmation and settlement platform
- Led requirements, analysis, and implementation of platform enhancements for high-volume trade processing
- Reduced application memory consumption by 30% through GC log analysis and systematic code refactoring

### Mahindra Comviva | Technical Manager | India | 2016 – 2017

- **Product:** PreTUPS - prepaid recharge distribution and mobile account management platform
- Directed product development, release management, and feature delivery across distributed teams
- Cut AWS cloud spending by 25% through strategic optimization and best practice implementation
- Improved application stability by 15% by identifying and resolving code inefficiencies

## R Systems International | Associate Technical Architect | India | US | 2015 – 2016

- **Client:** DMA Inc. Tax Interchange for Retail - enterprise point-of-sale tax solution
- Rebuilt client relationship and trust through consistent delivery and transparent communication
- Established architectural standards, design patterns, and code review frameworks
- Introduced Test-Driven Development (TDD), achieving 40% reduction in production defects
- Diagnosed and resolved performance bottlenecks using JVisualVM profiling tools

## JK Tech | Project Lead | India | UK | 2009 – 2015

- **Client:** Specsavers Optical Group - enterprise retail and supply chain systems
- Managed complete project lifecycles from initiation through post-implementation review
- Improved Agile delivery efficiency by 20% through process refinement and team coaching
- Promoted three times in six years: System Analyst → Technical Lead → Project Lead
- Recognized with "Star of the Iteration" awards (2012, 2013) and "Star of the Quarter" (June 2011)

## Fiserv | Senior Software Engineer | India | US | 2007 – 2009

- **Client:** Fiserv Trade Order Management - institutional trading platform
- Designed and built automated test framework executing 500+ tests in 7 minutes, saving 40% processing time
- Developed proof-of-concept solutions using emerging technologies (OpenESB, Apache ServiceMix)
- Delivered specifications, program changes, and comprehensive technical documentation

## JK Tech | Software Engineer | India | UK | 2005 – 2007

- **Internal Products:** iTimer timesheet application and corporate website modernization
- Introduced automated unit testing frameworks, reducing manual testing effort by one man-day monthly
- Mastered Agile and iterative development methodologies in production environments
- Supported QA, UAT, and production stabilization across multiple concurrent projects

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## AI Governance Systems and Safety Engineering

Built working systems across the control, evaluation, and defense layers of production AI.

**All code:** <https://github.com/sushilduseja>

### Governance Systems

**Constitutional Governance SDK** | Python, FastAPI, Groq, SQLite, Uvicorn, pytest

- Wraps any LLM provider with an asynchronous compliance layer that evaluates outputs against versioned plain-English rules without adding latency.
- Every evaluation writes to an append-only SQLite log with timestamp, model, score, and violated rule IDs.
- A live dashboard tracks compliance rates and violation trends per model.

### Enforcement Layer

**Code Safety Guardrails** | FastAPI, Groq, Python AST, SQLite, Redis, slowapi, pytest

- Intercepts LLM-generated Python before it reaches a runtime, running four validators in sequence: SQL injection, unsafe command execution, secrets exposure, and malicious imports.
- SQL vulnerabilities rewrite via AST to parameterized queries; blocked patterns return a structured denial with rule ID and matched text.

- The pipeline fails closed on parse errors and logs every request to SQLite.

**AI Code Reviewer** | Python, ThreadPoolExecutor, SIM.ai, pytest

- Three parallel agents assess security, performance, and code quality against git diffs simultaneously via ThreadPoolExecutor.
- Each agent returns findings with severity and line references; a report writer compiles them into structured markdown.
- Operates on any two commits with no configuration beyond an API key.

**Evaluation and Monitoring**

**FairLens Platform** | FastAPI, React, Vite, SQLite, Redis, JWT, Docker Compose, pytest

- Full-stack fairness audit tool that calculates demographic parity and disparate impact across protected attribute groups for financial ML models.
- A React frontend exposes results through JWT-authenticated dashboards; output is formatted for compliance submission.
- Runs locally or via Docker Compose.

**AI Guardian** | Python, Streamlit, Transformers, Plotly, pytest

- Defends LLM applications against prompt injection through four layered strategies: sanitization, pattern-based warning injection, prompt structure enforcement, and a composite mode combining all three.
- Each strategy logs attack success rate and confidence reduction per injection category.
- A Streamlit dashboard tracks per-strategy effectiveness over time across OpenAI, Anthropic, and local HuggingFace backends.

**FX Risk Monitor** | Java 21, Apache Kafka, Redpanda, Maven

- Event-driven FX trade risk system built on Kafka and Java 21 virtual threads.
- Producers publish trade events; consumers calculate risk scores and route high-risk alerts to a notification pipeline.
- A dead letter queue handles processing failures without dropping events..

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**Certifications & Executive Education**

**AI Strategy & Safety**

AGI Strategy, BlueDot Impact 2026  
Future of AI, BlueDot Impact 2026  
Generative AI for Executives 2024

**Cloud & Emerging Technology**

Cloud-Native & Multicloud Apps 2022  
Blockchain Essentials 2022

**Cybersecurity & Governance**

ISO/IEC 27001 Lead Auditor 2025  
AI Security & Governance 2025  
AI Security Fundamentals 2025  
Cybersecurity GRC 2024  
Foundations of Cybersecurity 2023

**Community & Thought Leadership**

AI Safety & Governance Community  
Active contributor · 2023-Present

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